

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory

Lab 6 – Test on C++ Pointers (Set A)

Date: 01.09.2026**Semester:** B.Tech 3rd Semester**Group:** 2**Instructions**

- Write all programs in C++.
- Use **pointers** wherever required in each question.
- Use pointer dereferencing and pointer arithmetic appropriately.
- Do not use array indexing where explicitly restricted.
- Properly release dynamically allocated memory using `delete` or `delete[]`.
- Display the output in a clear and readable format.

1. Delivery Counter

A delivery company stores the number of parcels delivered in a variable.

Write a C++ program that creates a pointer to this variable. Display the number of parcels using the pointer, increase the number by a value entered by the user using the pointer, and display the updated number.

2. Digital Wallet Balance

A digital wallet stores the current balance of a user.

Create a pointer pointing to the balance. Using the pointer:

1. Display the current balance.
2. Add a specified amount.
3. Deduct a specified amount.
4. Display the final balance.

3. Library Shelf

A library stores the identification numbers of 6 books in an array.

Write a program to:

1. Display all book IDs using a pointer.
2. Display the address of each book ID.

Condition: Traverse the array using pointer increment.

4. Cinema Seat Update

A cinema stores 8 seat numbers in an array.

Write a program that uses pointer arithmetic to change the seat number at a position entered by the user.

Display the seat numbers before and after the update.

Condition: Do not use `arr[position]` for updating the element.

5. Museum Visitor Update

A museum stores the number of visitors for the day.

Write a function:

```
void updateVisitors(int *count);
```

The function should accept the number of newly arrived visitors and update the original visitor count through the pointer.

Display the visitor count before and after calling the function.

6. Grocery Price Scanner

A grocery store stores the prices of 7 products.

Write a function that receives:

- A pointer to the first price.
- The number of products.

Using pointer traversal, find and display the highest price.

Condition: Do not use array indexing inside the function.

7. Message Inspector

A messaging application stores a sentence in a character array.

Using a character pointer, count:

- Number of uppercase letters.
- Number of lowercase letters.
- Number of spaces.

Traverse the sentence until the null character `'\0'`.

8. Game Score Adjustment

A game stores the scores of n players in an array.

Write a function that receives a pointer to the scores and the number of players. The function should increase every score by 10.

Display the scores before and after calling the function.

Condition: The original array must be modified using pointers.

9. Parking Slot Monitor

A parking system does not know in advance how many parking slots it needs to store.

Write a program that:

1. Dynamically allocates memory for n parking slot statuses.
2. Uses 0 for available and 1 for occupied.
3. Counts available and occupied slots using a pointer.
4. Releases the dynamically allocated memory.

10. Student ID Search

A university receives a variable number of student IDs.

Write a program that:

1. Dynamically allocates memory for n student IDs.
2. Accepts all student IDs.
3. Searches for a particular ID using pointer traversal.
4. Displays whether the ID is found and its position.
5. Properly deallocates the memory.

Condition: Do not use array indexing while searching.